ENTERPRISE CLOUD GOVERNANCE & PROVENANCE

Cloud Resource Archaeologist

Know why every cloud resource exists. Retire it safely with zero fear.

THE PROBLEM TODAY:

- Thousands of orphan servers, databases & disks running in company cloud accounts
- Nobody knows who created them or why they exist (tickets lost, employees left)
- Cleanup tools say 'Delete', but teams fear breaking production dependencies
- Result: Millions wasted annually on zombie cloud infrastructure

The Solution: Evidence-Based Answers

1. Why does it exist?
(PROVENANCE TIMELINE)

• Stitches evidence from AWS API, CloudTrail, Terraform & GitHub
• Builds chronological story from creation to last commit
• Flags manual drift changes outside CI/CD pipelines

2. Who owns it?
(OWNER RESOLUTION)

• Creator != Owner (creators leave or change teams)
• 5-tier fallback chain:
Tag -> CODEOWNERS -> Account Owner -> Creator -> Review
• Explicit HIGH / MEDIUM / LOW confidence rating

3. Can we delete it?
(RISK VERDICT)

• Evaluates dependencies, traffic history & IaC state
• Thin evidence safeguard: blocks verdict if clues < 2
• Never auto-deletes; requires human owner approval

The Differentiator: Reversible Quarantine

Eliminates the fear of decommissioning by making every step undoable.

1. EVIDENCE GATHERING: Correlate logs, usage, and IaC dependencies
2. OWNER APPROVAL: Present evidence & savings directly to verified team
3. PRE-DECOMMISSION SNAPSHOT: Full automated backup created first
4. ISOLATE / QUARANTINE: Apply deny-all security group or stop instance
5. 14-DAY OBSERVATION WINDOW: Monitor alarms, rejected calls, or user reports
6. 1-CLICK ROLLBACK OR DELETE: Restore in 3 minutes if needed, or permanently delete with full audit trail

MVP Scope & Business Impact

MVP SCOPE:
- AWS EC2, EBS Volumes & Security Groups
- Terraform State & GitHub PR attribution
- CloudTrail & CloudWatch metrics ingestion
- SQLite / PostgreSQL audit storage
- FastAPI REST engine & Streamlit Web UI
- Watsonx / ICA AI Agent interface

BUSINESS VALUE:
- 20% to 40% monthly cloud cost reduction
- Zero accidental outages via reversible rollback
- Hours saved per investigation (automated vs manual)
- Ground-truth feedback loop improves accuracy over time

STATUS: 100% Working Prototype Ready for Demo